

ROOKE'S CHARTS

Central crack in a rectangular sheet: uniform uniaxial tensile stress

A rectangular sheet of height $2h$ and width $2b$ contains a central crack of length $2a$; a uniform stress σ , perpendicular to the crack direction, acts over the ends of the sheet (see Fig.1). This configuration has been studied by Isida¹ using boundary collocation of complex stress functions. Curves of K_I/K_0 vs. a/b for various h/b ratios are shown in Fig.1; K_0 is the stress intensity factor for a crack in an infinite sheet ($h = b = \infty$) and is given by

$$K_0 = \sigma \sqrt{\pi a}. \quad (1)$$

The results for large values of h/b are in good agreement with those of other workers,^{2,3,4,5,6} Benthem and Koiter² have obtained a simple analytic expression by interpolating between the two limiting cases of small and large crack lengths,

$$\lim_{a/b \rightarrow 0} \left\{ \sqrt{1 - \frac{a}{b}} \frac{K_I}{K_0} \right\} = 1 - \frac{1}{2} \left(\frac{a}{b} \right) \quad (2)$$

and

$$\lim_{\substack{a/b \rightarrow \infty \\ h/b \rightarrow 1.0}} \left\{ \sqrt{1 - \frac{a}{b}} \frac{K_I}{K_0} \right\} = \frac{2}{\sqrt{\pi^2 - 4}}. \quad (3)$$

The interpolation formula which is accurate to 1 % for all a/b is

$$\frac{K_I}{K_0} = \frac{1 - 0.5 \frac{a}{b} + 0.326 \left(\frac{a}{b} \right)^2}{\sqrt{1 - \frac{a}{b}}}. \quad (4)$$

$\frac{1.3796}{2.9702}$

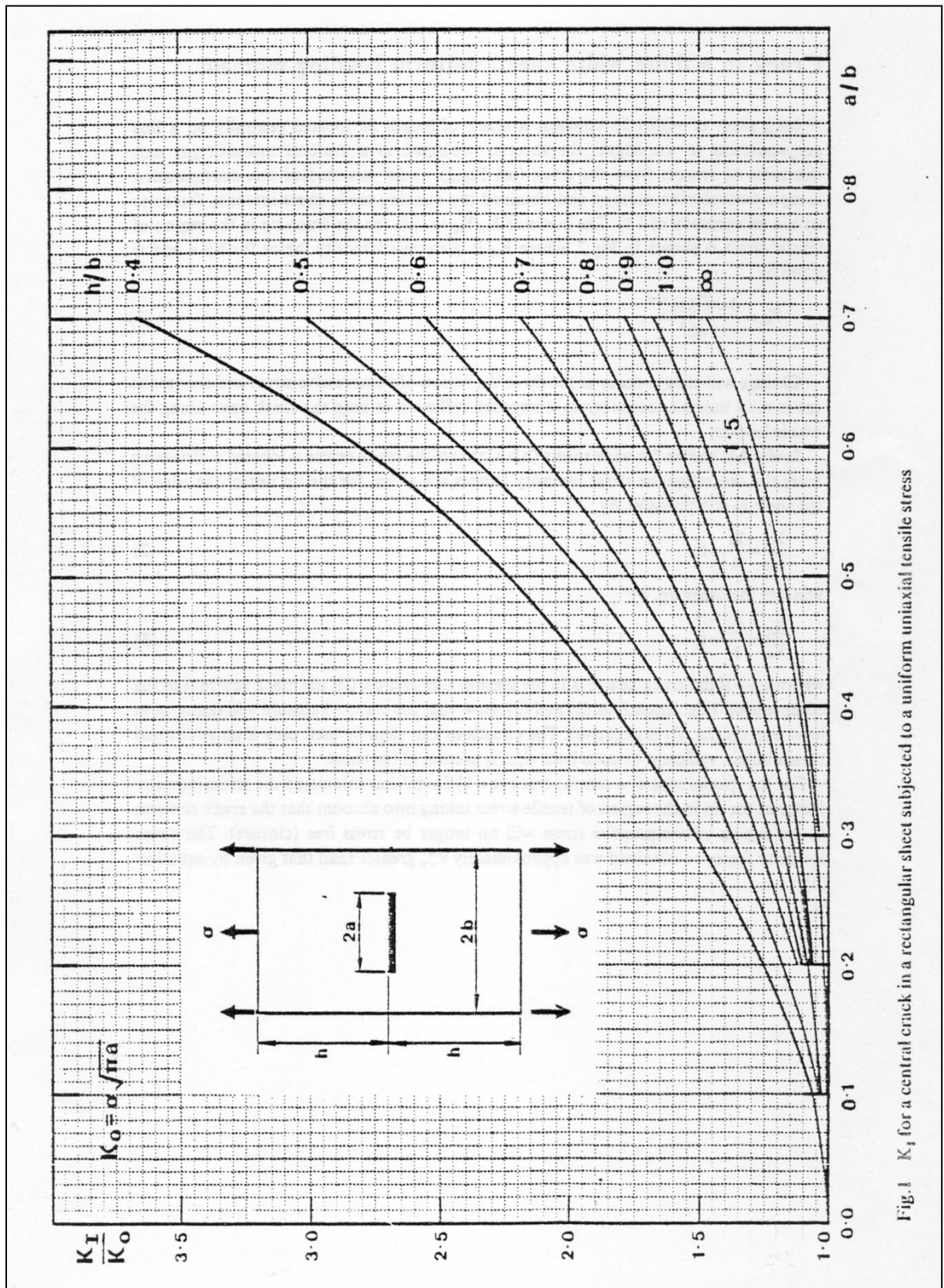


Fig.1 K_I for a central crack in a rectangular sheet subjected to a uniform uniaxial tensile stress

Edge crack in a finite width sheet: uniaxial tensile stress with or without bending constraints

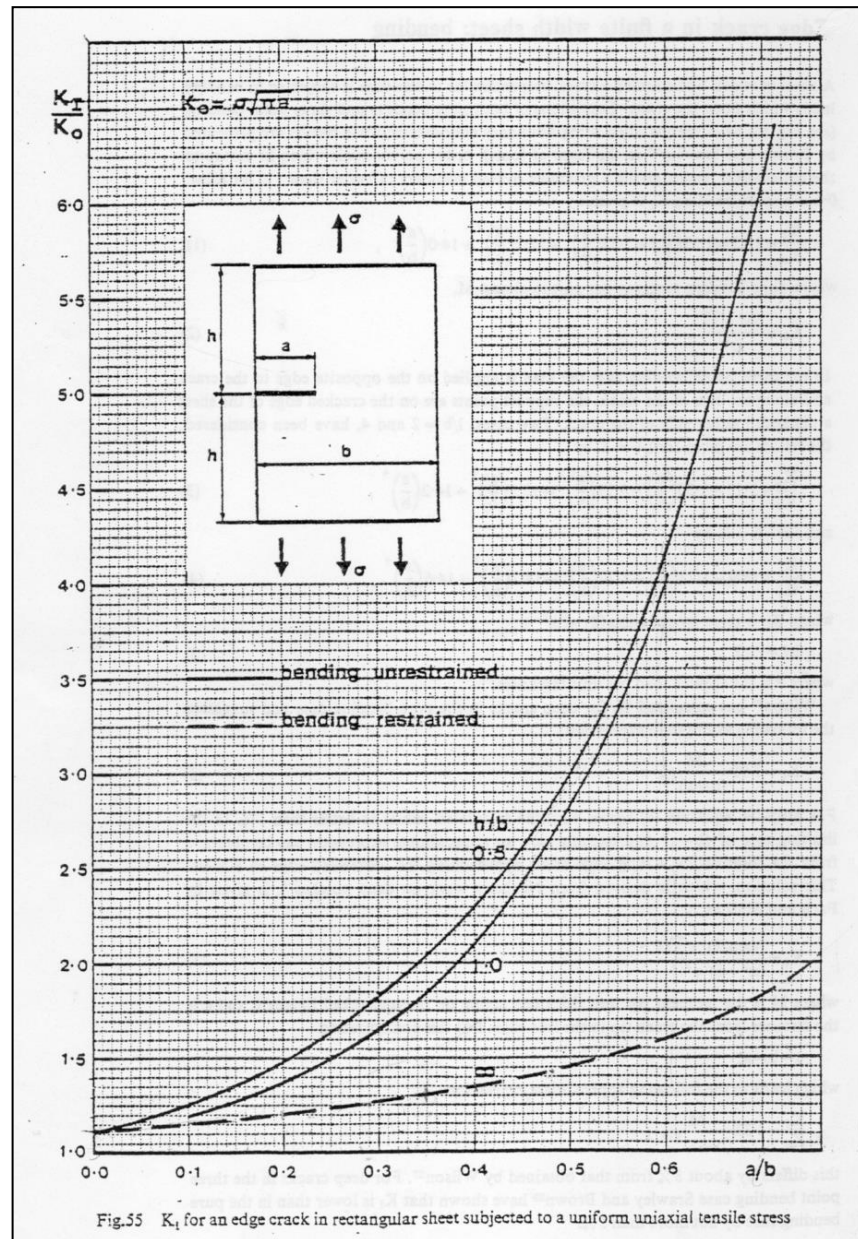


Fig.55 K_I for an edge crack in rectangular sheet subjected to a uniform uniaxial tensile stress

Rooke & Cartwright page. 84. (Fig.55)

Harris²⁵ has obtained results for an edge crack in a sheet which does not bend by considering a limiting case of a circumferential crack in a long cylinder (see 3.2.1). The results for K_I/K_0 are given by Harris as

$$\frac{K_I}{K_0} = \frac{5}{[20 - 13(a/b) - 7(a/b)^2]^{1/2}} \quad (4)$$

This equation, which is based on approximate solutions by Neuber²⁶ may be in error by up to 8%. Curves of K_I/K_0 vs. a/b for this case are also shown in Fig.55 for $h/b = \infty$.

Central crack in a rectangular sheet: forces at crack centre

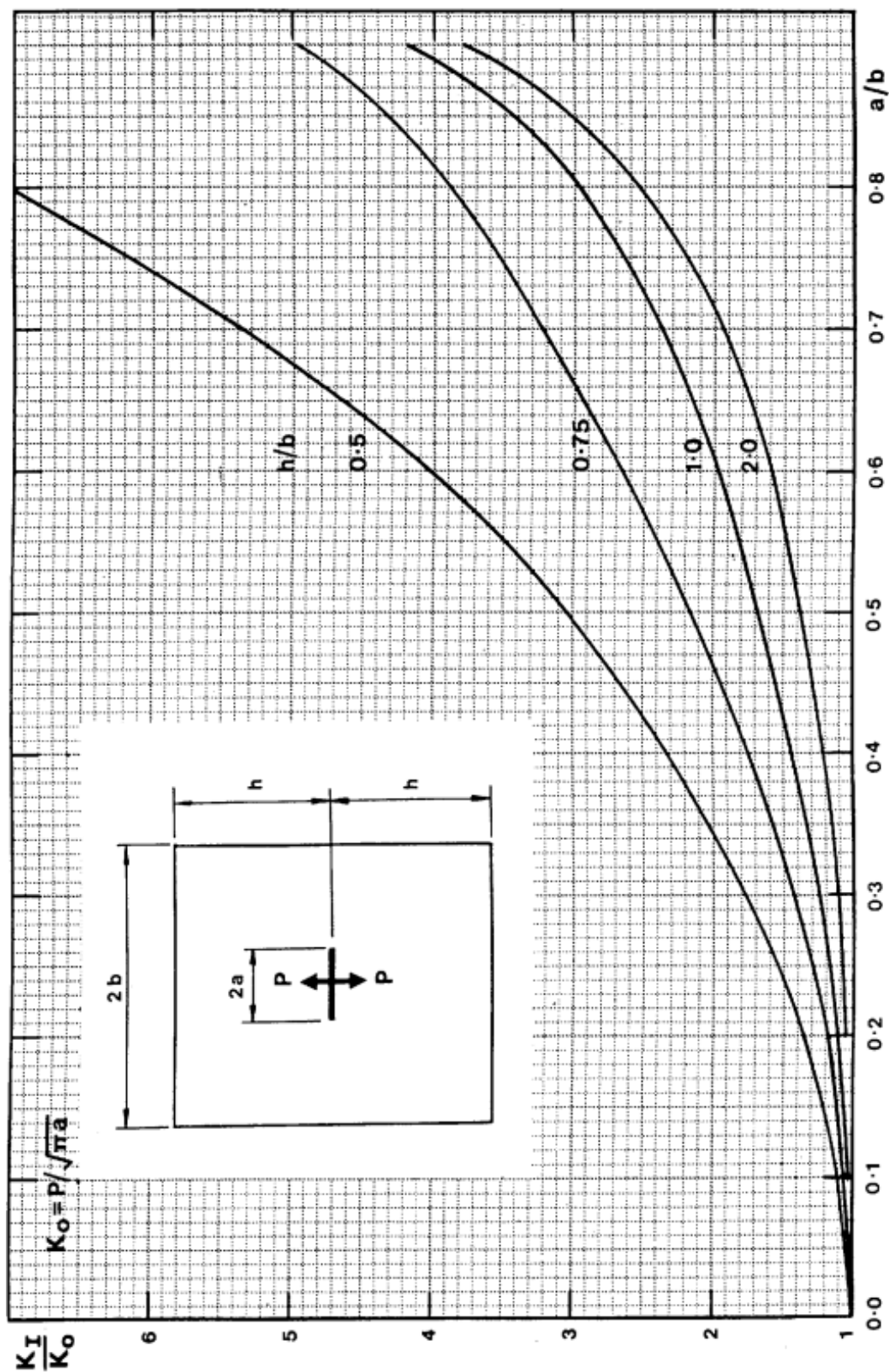


Fig.5 K_I for a central crack in a rectangular sheet with opposing forces at the centre of the crack

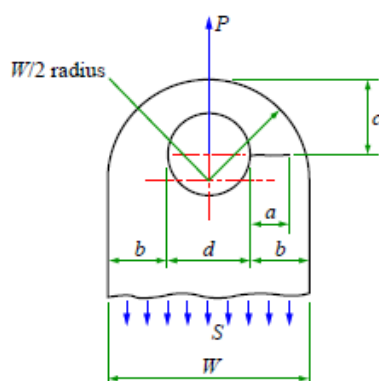
STRESS INTENSITY FACTORS IN LUGS (THROUGH-THICKNESS CRACKS)

1. NOTATION

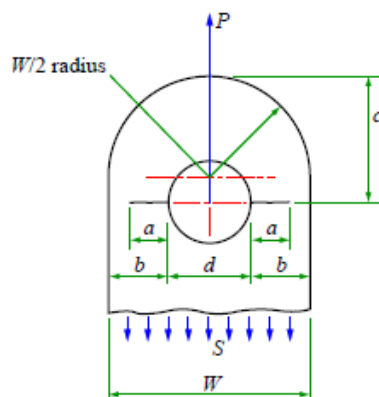
a	actual crack length as defined in Sketches 1.1 and 1.2	m	in
b	width of net section on one side of lug, thus $b = (W - d)/2$ (note that b is defined this way even when $c/W < 1/2$ as depicted in Sketch 1.1)	m	in
c	distance from centre of hole to lug end (see Sketches 1.1 and 1.2)	m	in
d	diameter of hole in lug	m	in
K_I	mode I stress intensity factor* defined by $K_I = S(\pi a)^{1/2} \alpha$	N/m ^{3/2}	lbf/in ^{3/2}
P	load on lug	N	lbf
S	stress based on gross cross-sectional area, $S = P/(Wt)$	N/m ²	lbf/in ²
t	thickness of lug	m	in
W	width of lug	m	in
α	geometrical factor* in K_I		

Both SI and British Units are quoted but any coherent system of units may be used.

* See ESDU 80036 "Introduction to the use of linear elastic fracture mechanics in estimating fatigue crack growth rates and residual strength of components" for a definition of mode I cracks and notes on α .



Sketch 1.1 Single crack



Sketch 1.2 Double cracks

TABLE 5.1 Values of α for Double Cracks Fully Through-Thickness in Round-Ended Lugs Axially Loaded by a Single Clearance-Fit Pin

c/W	d/W	a/b												
		0	0.025	0.05	0.1	0.15	0.2	0.25	0.3	0.4	0.5	0.6	0.7	0.8
0.4	0.3	6.84	6.03	5.49	4.69	4.02	3.60	3.31	3.10	2.80	2.63	2.54	2.57	2.74
	0.35	6.81	6.16	5.65	4.88	4.36	3.97	3.68	3.44	3.11	2.91	2.84	2.86	3.01
	0.40	7.02	6.47	6.01	5.31	4.79	4.40	4.09	3.88	3.49	3.28	3.17	3.17	3.32
	0.45	7.40	6.93	6.61	5.92	5.39	4.94	4.61	4.36	3.98	3.69	3.57	3.52	3.71
	0.55	8.44	8.10	7.77	7.19	6.71	6.30	5.98	5.68	5.17	4.81	4.60	4.53	4.68
0.5	0.22	7.09	5.88	5.10	4.03	3.39	2.95	2.65	2.49	2.24	2.14	2.10	2.18	2.38
	0.25	6.30	5.40	4.73	3.84	3.29	2.90	2.67	2.50	2.28	2.18	2.15	2.23	2.43
	0.27	6.07	5.28	4.70	3.87	3.33	2.98	2.72	2.56	2.33	2.23	2.21	2.28	2.49
	0.30	5.98	5.29	4.77	4.00	3.49	3.13	2.89	2.70	2.49	2.37	2.31	2.39	2.62
	0.40	6.23	5.76	5.39	4.79	4.31	3.96	3.70	3.50	3.22	3.08	3.04	3.11	3.37
	0.50	6.93	6.57	6.21	5.68	5.23	4.93	4.68	4.45	4.11	3.94	3.89	3.94	4.24
	0.55	7.42	7.07	6.78	6.29	5.90	5.59	5.32	5.09	4.73	4.53	4.42	4.48	4.77
	0.65	8.88	8.72	8.54	8.20	7.87	7.51	7.20	6.89	6.47	6.21	6.10	6.14	6.42
0.6	0.22	6.60	5.52	4.74	3.84	3.20	2.82	2.56	2.38	2.13	2.03	2.02	2.11	2.33
	0.25	6.30	5.37	4.70	3.86	3.26	2.90	2.62	2.46	2.23	2.15	2.15	2.23	2.47
	0.30	5.93	5.20	4.69	3.93	3.40	3.05	2.80	2.63	2.42	2.34	2.35	2.46	2.71
	0.35	5.76	5.18	4.76	4.10	3.63	3.30	3.07	2.89	2.68	2.59	2.59	2.71	3.00
	0.40	5.69	5.24	4.89	4.33	3.95	3.62	3.40	3.22	3.00	2.90	2.90	3.02	3.36
	0.45	5.77	5.42	5.12	4.63	4.29	4.00	3.78	3.61	3.37	3.25	3.25	3.40	3.73
	0.50	6.02	5.76	5.50	5.07	4.72	4.45	4.21	4.03	3.80	3.68	3.68	3.83	4.22
	0.55	6.52	6.26	6.00	5.58	5.24	4.97	4.74	4.50	4.33	4.21	4.19	4.33	4.75

TABLE 5.2 Values of α for Single Cracks Fully Through-Thickness in Round-Ended Lugs Axially Loaded by a Single Clearance-Fit Pin

c/W	d/W	a/b											
		0	0.025	0.05	0.1	0.15	0.2	0.3	0.4	0.5	0.6	0.7	0.8
0.4	0.3	6.84	5.99	5.32	4.39	3.74	3.30	2.72	2.36	2.11	1.99	1.92	1.98
	0.35	6.82	6.13	5.54	4.69	4.11	3.69	3.10	2.69	2.43	2.28	2.20	2.23
	0.40	7.02	6.40	5.91	5.11	4.58	4.11	3.50	3.06	2.78	2.59	2.50	2.55
	0.45	7.39	6.90	6.35	5.70	5.12	4.65	3.99	3.52	3.20	3.00	2.89	2.92
	0.55	8.44	8.09	7.75	7.10	6.53	6.02	5.30	4.73	4.30	4.00	3.83	3.93
0.5	0.22	7.09	5.65	4.80	3.76	3.10	2.65	2.12	1.82	1.66	1.58	1.55	1.60
	0.25	6.21	5.23	4.62	3.67	3.07	2.66	2.15	1.87	1.71	1.63	1.60	1.70
	0.27	6.01	5.20	4.59	3.71	3.12	2.69	2.19	1.94	1.78	1.70	1.67	1.77
	0.30	5.97	5.24	4.69	3.86	3.28	2.87	2.38	2.10	1.92	1.82	1.80	1.89
	0.40	6.23	5.73	5.32	4.60	4.10	3.70	3.13	2.82	2.60	2.49	2.47	2.58
	0.50	6.91	6.53	6.17	5.53	5.09	4.68	4.11	3.72	3.48	3.30	3.29	3.43
	0.55	7.42	7.09	6.75	6.18	5.72	5.31	4.74	4.33	4.05	3.88	3.82	3.97
	0.65	8.87	8.66	8.45	8.03	7.62	7.28	6.60	6.07	5.75	5.50	5.40	5.50
0.6	0.22	6.60	5.42	4.64	3.60	2.97	2.56	2.07	1.78	1.61	1.53	1.50	1.59
	0.25	6.25	5.28	4.60	3.63	3.07	2.68	2.17	1.90	1.73	1.65	1.64	1.73
	0.30	5.87	5.17	4.61	3.77	3.26	2.89	2.39	2.10	1.93	1.85	1.86	2.00
	0.35	5.70	5.17	4.69	3.98	3.49	3.13	2.63	2.36	2.18	2.09	2.09	2.24
	0.40	5.69	5.26	4.84	4.22	3.74	3.39	2.91	2.63	2.48	2.38	2.40	2.57
	0.45	5.78	5.45	5.11	4.57	4.10	3.78	3.30	3.00	2.82	2.74	2.76	2.96
	0.50	6.02	5.76	5.47	4.99	4.57	4.22	3.75	3.44	3.24	3.16	3.22	3.41
	0.55	6.54	6.26	5.97	5.50	5.10	4.77	4.29	3.97	3.77	3.68	3.82	3.95